

COMMANDER MISSION LOG: SOL 04 EXPEDITION UPDATE

1. Atmospheric sampling reveals surface barometric pressure at 6.1 mbar.
2. Regolith core drill completed at coordinates 18.4 N, 77.2 E.
3. Water ice sub-surface deposits detected at 1.4 meters depth.
4. Primary Stirling engine Kilopower reactor achieved criticality at 0400.
5. Coolant sodium-potassium loop flow rate stabilized at 1.82 kg/sec.
6. Habitat dome environmental life support pressure sealed at 101.3 kPa.
7. Oxygen recovery via MOXIE prototype yielded 12.4 grams per hour.
8. Crew physical conditioning: all 6 astronauts within nominal bounds.
9. Emergency rover EV-1 battery pack recharge cycle verified via solar B.
10. Communication latency to Earth DSN measured at 14.2 minutes.

FIELD NOTE SIGN-OFF:

Dust deposition on exterior radiator panels exceeded baseline predictions
by 14 percent following local convective dust devil.
Mechanical wiper sweeps scheduled every 48 hours to preserve cooling.